

MINI HACKATHON 2026

SentinelShield AI

*An Autonomous AI-Powered Financial Security Operations Center
for Intelligent Fraud Detection, Risk Intelligence & Cyber Threat Response*

PROJECT PROPOSAL

Round 1 — Category: AI in Finance & Cybersecurity

Field	Details
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Round	Round 1 — First Round

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1. Executive Summary

The global banking sector is undergoing rapid digital transformation, and with it, financial institutions face an expanding attack surface across mobile banking, digital wallets, and real-time payment rails. Conventional, rule-based fraud controls struggle to keep pace with adaptive attackers, resulting in delayed detection, high false-positive rates, and limited transparency for both security analysts and end customers.

SentinelShield AI is an autonomous, AI-powered Financial Security Operations Center (Financial SOC) that unifies transaction monitoring, identity and access risk analysis, and behavioral threat detection into a single, explainable platform. It combines supervised machine learning, rule-augmented heuristics, and Explainable AI (XAI) to convert every login and transaction event into a dynamic, evidence-backed risk score, and to trigger a proportionate, automated response — from passive logging to account freezing.

The proposed system is built on a modern, cloud-ready technology stack (React/TypeScript, Spring Boot, FastAPI, PostgreSQL) and has already been implemented as a working full-stack prototype, including a customer security portal, an administrator SOC dashboard, a weighted AI Risk Engine, and a live Cyber Attack Simulator that demonstrates the complete detect-analyze-respond pipeline. This proposal presents the industry context, technical architecture, functional scope, innovation rationale, and commercialization pathway for the project, in alignment with the Mini Hackathon 2026 judging criteria of Technology, Innovation, Concept, Feasibility, and Presentation.

2. Introduction

2.1 Digital Transformation in Banking

Banks and fintech providers have shifted decisively toward digital-first channels — mobile applications, API-driven open banking, and instant payment networks — which has significantly increased both convenience and the volume of exploitable digital touchpoints available to threat actors.

2.2 The Rise of Financial Cybercrime

Financial services remain among the most heavily targeted sectors for cybercrime, spanning organized fraud rings, credential-theft campaigns, and insider misuse. Attackers increasingly use automation, distributed infrastructure, and social engineering to bypass static, signature-based defenses.

2.3 Importance of AI in Cybersecurity

Artificial intelligence enables adaptive, pattern-based threat detection that generalizes beyond hand-written rules, using techniques such as anomaly detection, behavioral profiling, and ensemble classification to identify novel and evolving attack patterns with greater precision than static rule engines.

2.4 The Need for a Financial SOC

A Financial Security Operations Center consolidates monitoring, detection, scoring, explanation, and response into one operational layer, giving institutions the equivalent of an enterprise Security Orchestration, Automation and Response (SOAR) capability, purpose-built for financial risk and fraud use cases.

3. Problem Statement

Financial institutions and fintech platforms continue to face escalating exposure across several recurring threat categories:

- Fraudulent and anomalous transactions that bypass static velocity and threshold rules
- Account Takeover (ATO) via credential theft, impossible-travel logins, and session hijacking
- Credential stuffing and brute-force attacks against authentication endpoints
- Phishing and look-alike domains that harvest customer credentials
- Insider threats involving abnormal data access by privileged users
- Automated bot activity abusing banking and payment interfaces

These threats are compounded by a structural weakness in traditional systems: rule-based detection engines are reactive by design, apply uniform thresholds regardless of context, and generate binary outputs with little supporting rationale. This limits an analyst's ability to triage quickly, produces high false-positive volumes that erode customer trust, and leaves institutions exposed during the critical window between compromise and detection.

There is a clear need for an intelligent, explainable, AI-powered Financial Security Operations Center capable of detecting and responding to cyber threats in real time.

4. Proposed Solution

SentinelShield AI addresses these gaps through a centralized, AI-driven risk analysis layer that intercepts every authentication and transaction event before it is finalized.

4.1 Objectives

- Detect known and emerging fraud patterns using both supervised ML and behavioral heuristics
- Produce a transparent, human-readable rationale for every AI decision (Explainable AI)
- Automate proportionate incident response without unnecessary friction for legitimate users
- Provide real-time visibility to both customers and security analysts

4.2 High-Level Workflow



Figure 1: SentinelShield AI — high-level solution workflow

4.3 Expected Benefits

- Reduced Mean Time to Detect (MTTD) and Mean Time to Respond (MTTR) for financial threats
- Lower false-positive rates through weighted, multi-signal risk scoring rather than single-rule triggers
- Increased customer trust via transparent, explainable fraud decisions
- Operational efficiency for SOC analysts through a consolidated Incident Center and automated triage

5. System Architecture

SentinelShield AI follows a layered, service-oriented architecture that separates the presentation, business logic, AI inference, and persistence concerns — enabling each layer to scale and evolve independently.

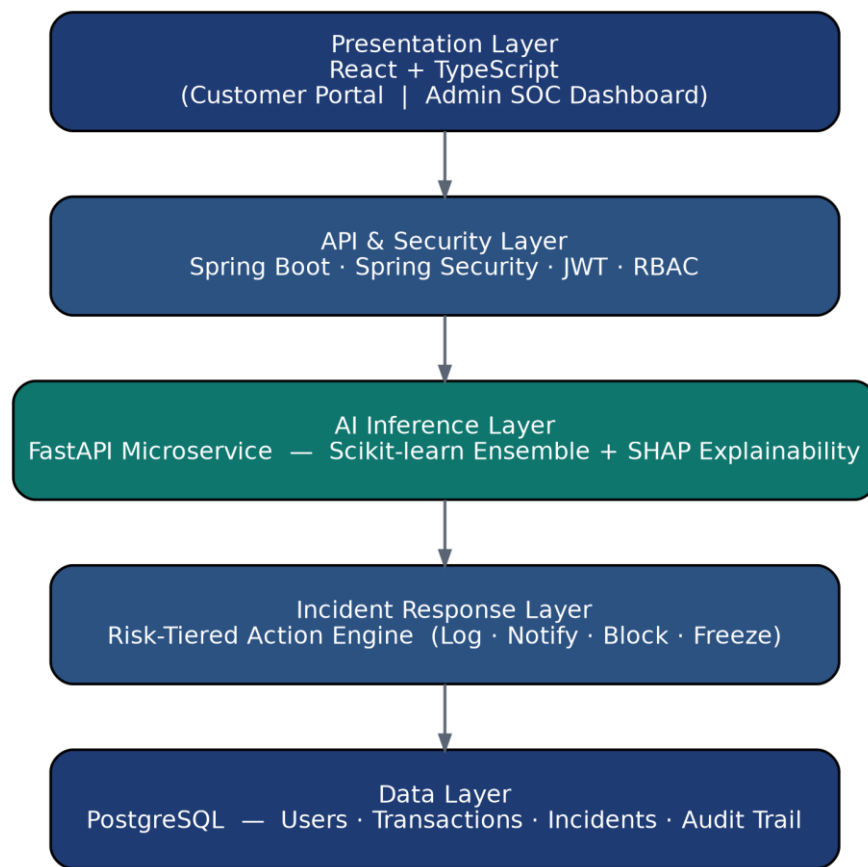


Figure 2: Enterprise system architecture — layered service design

Communication between the frontend and backend occurs over authenticated, HTTPS-secured REST APIs. The backend forwards relevant login and transaction events to the AI microservice for scoring; the resulting risk score, confidence level, and explanation are persisted to the audit trail and pushed to the relevant dashboard in real time, ensuring traceability for compliance and post-incident review.

6. Core Functional Modules

6.1 Customer Portal

- Personal Security Dashboard with an aggregated security score
- Transaction history with detailed, drill-down views
- Login history and trusted-device management (device fingerprinting)
- Real-time fraud and security alerts
- Self-service suspicious-activity reporting

6.2 Admin SOC Dashboard

- Live transaction and authentication monitoring across the user base
- Incident Center — centralized case management with severity and status tracking
- Risk analytics: daily risk trend lines, attack-type distribution, high-risk user cohorts
- Threat intelligence view: top offending countries, IP ranges, and repeat-offender accounts
- Account controls: suspend, unlock, and freeze, with a full audit trail
- Automated and on-demand compliance/incident reporting

6.3 AI Risk Engine

A weighted, multi-signal scoring engine that fuses transactional, behavioral, and contextual features (amount, geo-velocity, device fingerprint, network reputation, authentication history) into a single dynamic risk score per event.

6.4 Explainable AI (XAI) Layer

Every AI decision is accompanied by a structured, human-readable rationale — the specific signals that drove the score — rather than an opaque binary classification, supporting faster analyst triage and stronger customer communication.

6.5 Cyber Attack Simulator

An interactive module that lets administrators trigger realistic attack scenarios (e.g., a high-value transaction from an unrecognized device on a foreign VPN) and observe the full detect–score–explain–respond pipeline execute live, turning the platform into a demonstrable, judge-facing showcase rather than a static pitch.

7. Incident Detection Framework

The platform maps each threat category to a specific detection technique and a corresponding automated response, forming a structured Incident Detection Matrix:

Incident	Detection Technique	AI / System Response
High-Value / Anomalous Transaction	ML Classification (ensemble model)	Block transaction, alert analyst
Impossible Travel	Geo-velocity analysis	Require step-up MFA
VPN / Anonymized Login	Threat intelligence & IP reputation	Notify user, increase risk score
Unknown Device	Device fingerprinting	Increase risk score, request verification
Repeated Failed Login	Rule-based threshold engine	Temporarily lock account
Credential Stuffing	Behavioral analytics (UEBA)	Block source IP
Insider Threat	User & Entity Behavior Analytics	Alert security analyst
Phishing URL	NLP + URL/domain similarity analysis	Block URL, notify user

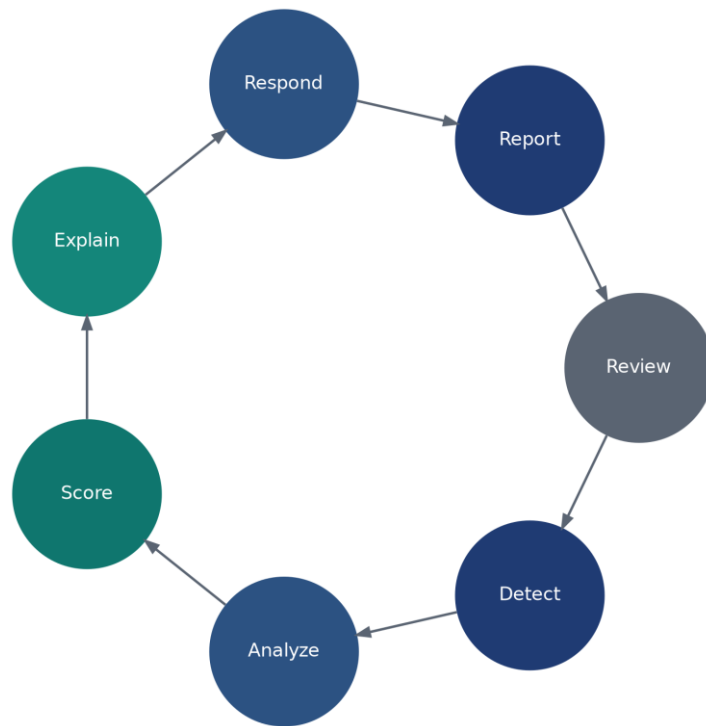


Figure 3: Incident response lifecycle

8. AI & Machine Learning Approach

The AI Risk Engine is designed around an ensemble of complementary techniques, balancing detection accuracy, interpretability, and resilience to novel (previously unseen) attack patterns:

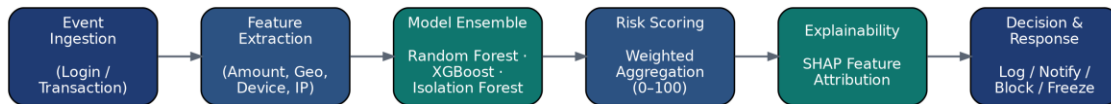


Figure 4: AI processing pipeline — from event ingestion to automated response

Technique	Role in the Pipeline
Random Forest	Baseline supervised classifier for structured transaction features
XGBoost	Gradient-boosted ensemble for higher-precision fraud classification
Isolation Forest	Unsupervised anomaly detection for previously unseen fraud patterns
SHAP (SHapley Additive exPlanations)	Feature-level attribution powering the Explainable AI layer
Behavioral / UEBA Analytics	Establishes per-user behavioral baselines to flag deviations
Weighted Risk Scoring	Aggregates model outputs and rule signals into one actionable score

Model performance will be evaluated using standard classification metrics — precision, recall, F1-score, and ROC-AUC — with particular emphasis on minimizing false positives, since excessive false alerts directly degrade customer experience and analyst trust in the system.

9. Technology Stack

Category	Technologies
Frontend	React, TypeScript, Vite, Tailwind CSS, Shaden UI, Framer Motion, Recharts, Axios, React Router
Backend	Spring Boot, Spring Security, JWT, REST APIs
AI Service	Python, FastAPI, Scikit-learn, SHAP
Database	PostgreSQL
Security	BCrypt password hashing, RBAC, HTTPS/TLS, input validation
Development Tools	Git/GitHub, VS Code, Postman
Deployment (Planned)	Docker, CI/CD (GitHub Actions), cloud hosting

10. Cybersecurity Features

Feature	Purpose
JWT Authentication	Stateless, tamper-resistant session tokens for API access
Role-Based Access Control (RBAC)	Restricts customer vs. administrator capabilities
BCrypt Password Hashing	One-way, salted password storage resistant to brute force
Audit Logging	Immutable trail of security-relevant actions for compliance
Input Validation	Mitigates injection and malformed-request vulnerabilities
API Security	Authenticated, rate-limited, and schema-validated endpoints
Secure Session Management	Token expiry, refresh, and revocation controls
HTTPS / TLS	Encrypts data in transit between client and server
Risk Intelligence Feeds (Planned)	External IP/domain reputation to enrich scoring
Automated Incident Response	Rule-tiered actions from logging through account freeze

11. Innovation

SentinelShield AI's core innovation lies in shifting financial security from a static, rule-based posture to an adaptive, explainable, and largely autonomous one:

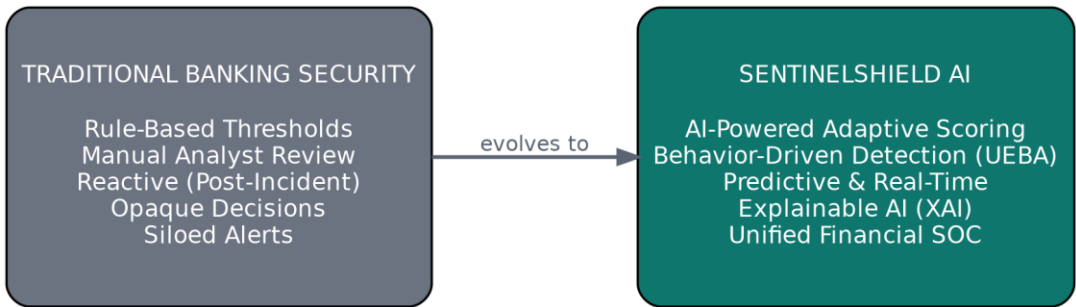


Figure 5: Traditional banking security vs. SentinelShield AI

This positions the platform closer to modern Zero Trust and adaptive-security principles — continuously verifying risk on every event rather than trusting a session by default — while remaining lightweight enough to integrate into an existing digital banking stack.

12. Commercialization Strategy

12.1 Target Market

Segment	Description
Commercial & Retail Banks	Modernizing legacy, rule-based fraud operations
Digital / Neobanks	Mobile-first institutions needing built-in SOC capability
FinTech & Payment Gateways	High transaction volume, limited in-house security teams
Insurance & Credit Providers	Secondary market for identity and claims-fraud risk scoring

12.2 Revenue Model

Revenue Stream	Description
SaaS Subscription	Tiered pricing by monitored accounts or transaction volume
Enterprise Licensing	On-premise / private-cloud deployment for regulated institutions
API-as-a-Service	Pay-per-call risk scoring API for smaller fintech integrators
Sandbox / Free Tier	Limited free access to drive fintech-startup adoption

13. Feasibility Analysis

13.1 Technical Feasibility

The stack relies entirely on mature, well-documented technologies (React, Spring Boot, FastAPI, PostgreSQL, Scikit-learn) with a working prototype already implemented, substantially de-risking further development.

13.2 Business Feasibility

The tiered SaaS and API-as-a-Service model allows adoption to start small (a single fintech pilot) before scaling to enterprise licensing, aligning cost with institution size.

13.3 Operational Feasibility

The modular architecture (frontend, backend, AI service) allows the four-member team to develop, test, and demo each component independently and in parallel.

13.4 Scalability

The AI inference layer is decoupled as its own microservice, allowing it to be scaled or replaced independently of the core backend as transaction volume grows.

13.5 Security & Compliance

JWT-based authentication, RBAC, encrypted transport, and immutable audit logging provide a foundation consistent with financial-sector expectations such as PCI-DSS-aligned controls and data-protection regulations, to be formally assessed ahead of any production pilot.

13.6 Cost Feasibility

Built entirely on open-source frameworks and libraries, the platform minimizes licensing costs during development, with cloud hosting the primary recurring cost at production scale.

14. Team Responsibilities

Member	Primary Responsibilities	Key Technologies
Shehani Lakshika Chandrakumar (Leader) — IT24104075	Project coordination, security architecture, backend authentication & integration	Spring Boot, Spring Security, JWT
Parameswaran Mayureshan — IT24100024	AI Risk Engine design, model development, risk-scoring logic	Python, FastAPI, Scikit-learn
Ravichandran Harini — IT24103952	Frontend development, customer & admin dashboard UI/UX	React, TypeScript, Tailwind CSS
Lafry A.F.H. — IT24103834	Testing & QA, documentation, presentation and demo preparation	Postman, Recharts, PowerPoint

Role assignments reflect the current development split and remain flexible as the team iterates through subsequent rounds.

15. Development Roadmap

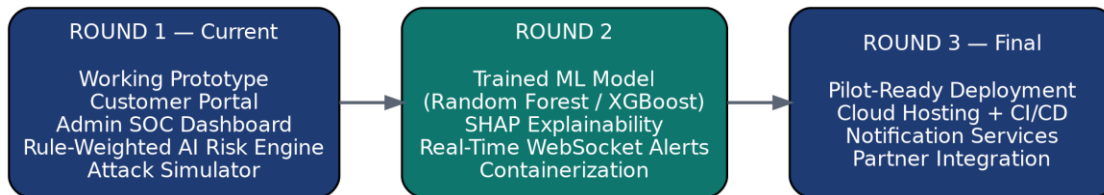


Figure 6: Development roadmap — Round 1 through final product

16. Future Scope

- AI copilot for SOC analysts (natural-language incident summarization)
- Containerized, cloud-native deployment with Docker and Kubernetes
- Event streaming and caching with Kafka and Redis for higher throughput
- Integration with external threat intelligence feeds
- Zero Trust access controls across all service-to-service communication
- Federated learning to improve fraud models across institutions without sharing raw data
- Blockchain-backed, tamper-evident audit logs for regulatory assurance

17. Conclusion

SentinelShield AI reframes financial fraud defense as a unified, explainable, and largely autonomous Financial Security Operations Center — combining ensemble machine learning, behavioral analytics, and transparent decision-making with automated, risk-tiered incident response. With a working full-stack prototype already in place, Team Quantix is well positioned to advance the platform's AI accuracy, security hardening, and enterprise scalability through the remaining rounds of Mini Hackathon 2026, and ultimately toward a pilot-ready product for real financial institutions.